University of Houston

ECE - 6370

Lab-3

User Manual

Hardware-Based True Random Number Generator and Timer Control System on FPGA

Under the Supervision

of

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1. Overview

This manual explains how to operate the FPGA-based single-player game system designed for Lab 3. The system uses a password-protected access mechanism, a hardware-based True Random Number Generator (RNG), and a two-digit countdown timer. The player attempts to match a randomly generated 4-bit number before the timer expires. The game is defined in such a way that it can only be played after entering the correct password. Once you are in the game the timer starts from 99->00 decrements like a timer.

The systems include:

- Password authentication
- A random number generator (0–15)
- A player-controlled 4-bit input
- A countdown timer starting at 99 seconds
- Game start and restart logic
- LED indicators for authentication, matching, and timeout
- Scoring
- Logout Function: Allows the user to exit the authenticated state when the timer is idle by pressing the LOAD button

2. Required Hardware

- Intel/Altera FPGA development board
- USB cable
- Power Supply

3. System Components

a) Authentication Inputs

- Password Switches:
  Leftmost 4 switches (SW9-SW6)

- ENTER / GAME Button:
  Used for password entry, game start, and game restart
  Buttons Used: *KEY 1*

- Logged-In LED:
  Turns ON after successful authentication
  LED Used: *LEDR0*

b) Random Number Generator (RNG)

- **RNG Button:** *KEY 3*
- While held, a 4-bit counter increments at 50 MHz
- When released, the final value is latched
- RNG output is shown on the display

c) Player Input

- **Player Switches:** 4-bit number selected using switches *(SW3-SW0)*
- **LOAD Button:** Loads the player's number **Button used:** *KEY 2*
- **Player Display:** Shows the loaded 4-bit value

d) Countdown Timer

- Displays remaining time in base-10
- Loads "99" after authentication
- Starts counting down when the GAME button is pressed
- Stops at "00" and blocks further input

e) LED Indicators

- **Logged-In LED:** LED R0
- **Matching LED:** LED R8
- **Non-Matching LED:** LED R9

f) Scoring Feature: The scoring has been implemented

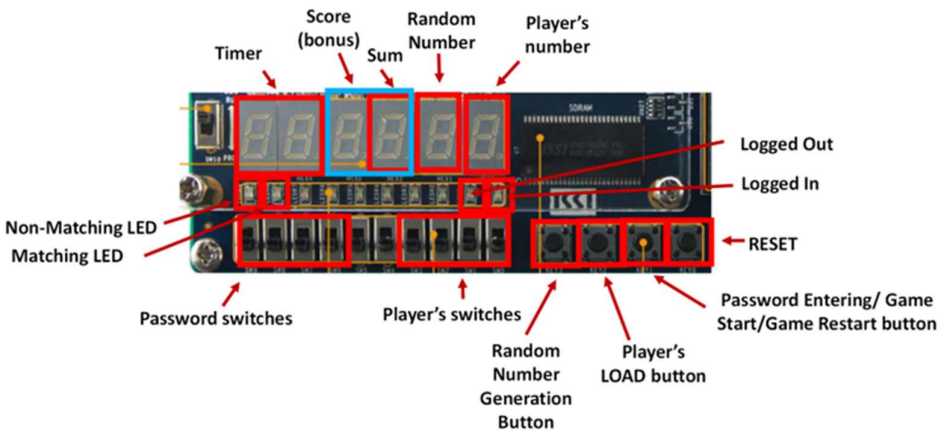
g) Logout Function

The **LOAD button (KEY2)** also serves as a **Logout button** when the timer is idle. If the user is authenticated **but the game is not running**, pressing LOAD logs the user out. After logout:

- Logged-In LED turns OFF
- Logged-Out LED turns ON
- All gameplay inputs (RNG, LOAD, timer) are disabled
- The system returns to password entry mode

This feature allows the user to exit the authenticated state without resetting the entire FPGA.

4. Board Layout



5. How to Authenticate System

i. Turn ON the FPGA board.
ii. Use the leftmost 4 switches to enter the last four digits of your password (UH ID ex: '1', '3', '8', '8').
iii. Press the ENTER button once.
iv. Repeat for all four digits.
v. When the password is correct, the Logged-In LED turns ON.
vi. The timer automatically loads **99** seconds.
vii. The system is now unlocked and ready to play.
viii. If you wish to exit the authenticated state before starting the game, press the LOAD button (KEY2) to log out.

- Logged-In LED turns OFF
- System returns to password entry mode

6. How to Play the Game

After the system is authenticated, the GAME button begins the round. Pressing this button loads the countdown timer with **99 seconds** and starts the countdown. Until the timer is running, both the Random Number Generator (RNG) button and the LOAD button remain disabled to prevent any inputs before the game officially begins.

Once the timer is active, a random number is generated by pressing and holding the RNG button. While the button is held, an internal 4-bit counter increments rapidly; when the button is released, the final counter value is latched and shown on the RNG display. The player then selects a 4-bit number using the designated switches and presses the LOAD button to store it. The Player display updates to show the loaded value.

The system automatically adds the Player's loaded number and the RNG value. If the **sum equals F (1111_2)**, the Matching LED turns ON and the system increments the score counter. If the sum is anything other than F, the Non-Matching LED turns ON and the score remains unchanged. The player may continue generating new random numbers and loading new values as long as time remains on the countdown.

When the timer reaches **00**, the Time-Up LED turns ON, and both the RNG and LOAD buttons are disabled so no further inputs can be made. At this point, the system displays the final score on the designated score displays, showing how many successful matches were achieved during the round.

To begin a new round, press the ENTER/GAME button again to reload the timer and restart gameplay.

Logout Option: If the system is authenticated but the timer has not started yet, pressing the LOAD button logs the user out and returns the system to password entry mode.

7. Expected Behavior

- RNG only works after authentication and before timeout.
- Timer displays "00" before authentication.
- Timer loads "99" after authentication.
- Timer decrements once per second.
- Timer stops at "00" and blocks all inputs.
- LEDs always reflect the correct system state.
- Score increments only when the sum of RNG + Player value equals **F**.
- **Logout function:** If authenticated but the timer has not started, pressing LOAD logs the user out and returns the system to password entry mode.
- System uses an LFSR-based RNG instead of a binary counter.
- Access control is handled by two FSMs (access_fsm and timer_fsm) for cleaner modular behavior.

8. Conclusion

This Lab 4 system expands the original design by integrating ROM-based authentication, an LFSR-based random number generator, a modular two-FSM access controller, and a logout feature. The result is a more secure, efficient, and hardware-realistic FPGA implementation. The system remains stable and interactive, allowing a single player to complete as many rounds as possible before time expires, while offering improved internal structure, clearer control flow, and enhanced user functionality.